

# Seok-Jin Kang

[laughingkang@korea.ac.kr](mailto:laughingkang@korea.ac.kr) | [GitHub](#) | [Website](#) | [Google Scholar](#)

## Bibliography

**Seok-Jin Kang, Ph.D.** is an **immunologist** and **computational biologist** whose academic and research trajectory bridges traditional immunology and modern **data-driven biology**. During his M.S. and Ph.D. training in the **Immune Modulation Laboratory** (PI: Prof. Taehoon Chun) at **Korea University**, he investigated **immune regulation**, **macrophage polarization**, and the **molecular mechanisms** of **host-pathogen interactions** using both *in vitro* and *in vivo* models.

He strategically pivoted from wet-lab immunology to computational biology to overcome the limitations of experimental methods in capturing a systems-level understanding of complex immune networks. Recent work focuses on the **modeling and identification of intrinsically disordered regions (IDRs)** using **quantum computing**, as well as applying **computer-aided drug design (CADD)** to uncover the **biophysical mechanisms** underlying **protein function and regulation**.

In the short term, his research aims to advance **precision medicine** through the development of **multi-omics-based computational models** capable of predicting **immune responses**, guiding **therapeutic design**, and enabling **personalized intervention strategies**. Over the long term, he aspires to establish himself as an **independent faculty member** and contribute to extending **human healthspan and wellbeing** through integrative, **AI-driven biomedical research** that unites **immunology**, **genomics**, and **computational modeling**.

## Research Interests

Precision Medicine, CAR-T Cell Therapy, Multi-omics Integration, Computer-aided Drug Design (CADD)

## Careers

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|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Korea University</b> (2026 QS ranking: 61 <sup>st</sup> )<br><i>Research Professor; Immune Modulation Laboratory (Advisor: Prof. Taehoon Chun)</i> | <b>Mar 2026 – Present</b>  |
| <b>Korea University</b><br><i>Postdoctoral Fellow, Immune Modulation Laboratory (Advisor: Prof. Taehoon Chun)</i>                                     | <b>Mar 2025 – Feb 2026</b> |

## Education

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Korea University</b> (2026 QS ranking: 61 <sup>st</sup> )<br><i>M.S. &amp; Ph.D. in Biotechnology</i><br><i>Immune Modulation Laboratory (Advisor: Prof. Taehoon Chun)</i> | <b>Mar 2018 – Feb 2025</b> |
| <b>Korea University</b><br><i>B.S. in Biotechnology</i>                                                                                                                       | <b>Mar 2014 – Feb 2018</b> |

## Publications

† indicates first author; \* indicates corresponding author.

### Featured Publications

5. **SJ Kang†**, H Shin\*, Biophysical mechanisms of spider-silk constituting element-induced stick-slip behavior and hydrogen bond regeneration for high toughness in silk fibers, *International Journal of Biological Macromolecules*, 147027, 2025 (Impact factor : 8.5, Q1)
4. **SJ Kang†**, H Shin\*, Amino acid sequence-based IDR classification using ensemble machine learning and quantum neural networks, *Computational Biology and Chemistry*, 108480, 2025 (Impact factor : 3.1, Q1)
3. **SJ Kang†**, J Yang†, NY Lee, CH Lee, IB Park, SW Park, HJ Lee, HW Park, HS Yun, T Chun\*, Monitoring cellular immune responses after consumption of selected probiotics in immunocompromised mice, *Food Science of Animal Resources* 42 (5), 903, 2022 (Impact factor : 3.7, Q2)

2. **SJ Kang**†, IB Park, T Chun\*, Open reading frame 5 protein of porcine circovirus type 2 induces RNF128 (GRAIL) which inhibits mRNA transcription of interferon- $\beta$  in porcine epithelial cells, *Research in Veterinary Science* 140, 79–82, 2021 (Impact factor : 1.8, Q2)
1. **SJ Kang**†, T Chun\*, Structural heterogeneity of the mammalian polycomb repressor complex in immune regulation, *Experimental & Molecular Medicine* 52 (7), 1004–1015, 2020 (Impact factor : 12.9, Q1)

#### Under Review

- **SJ Kang**†, H Shin\*, Quantum-Enhanced Weighted Gene Co-expression Network Analysis Reveals Regulatory Networks Underlying Sexual Size Dimorphism in *Macrobrachium nipponense*, *Integrative Zoology*, under review (Round 1), 2026 (Impact factor : 3.7, Q1)
- **SJ Kang**†, H Shin\*, Quantum-Enhanced Transfer Learning for IDR Binding Partner Prediction, *IEEE Transactions on Computational Biology and Bioinformatics*, under review, 2025 (Impact factor : 3.4, Q1)

#### Published Articles

- SP Choi†, J Yang†, IB Park†, **SJ Kang**, SW Park, CH Lee, HJ Lee, J Bae, CY Choi, J Shin, JI Kim, HY Jin, YS Lee, T Chun\*, Targeting MAFB potentiates immune checkpoint inhibitor efficacy by reprogramming tumor-associated macrophages to an M1-like phenotype in colorectal cancer, *Translational Research*, 13, S1931-5224(26)0042-3, 2026 (Impact factor : 5.9, Q1)
- SW Park†, CY Choi†, IB Park†, **SJ Kang**, HJ Lee, JI Kim, J Bae, D Geum, YP Cheon, T Chun\*, LDB1 represses fetal hemoglobin expression by enhancing BCL11A transcription, *Redox Biology*, 118, 108480, 2026 (Impact factor : 11.9, Q1)
- JH Yoon†, GB Yeon†, H Lee, H An, J Oh, **SJ Kang**, IB Park, SA Lim, SS Hwang, DS Kim, JH Kim, T Chun, YX Fu, J Bae\*, Tumor-targeted delivery of CXCL10 by mesenchymal stromal cells potentiates adoptive T cell therapy to treat solid tumors, *Biomedicine & Pharmacotherapy* 192, 118579, 2025 (Impact factor : 7.5, Q1)
- CH Lee†, HJ Lee†, SW Park, J Shin, **SJ Kang**, IB Park, HK Kim, T Chun\*, Mutational analysis of pig tissue factor pathway inhibitor  $\alpha$  to increase anti-coagulation activity in pig-to-human xenotransplantation, *Biotechnology Letters* 46 (4), 521–530, 2024 (Impact factor : 2.1, Q3)
- SW Park†, IB Park†, **SJ Kang**, J Bae, T Chun\*, Interaction between host cell proteins and open reading frames of porcine circovirus type 2, *Journal of Animal Science and Technology* 65 (4), 698, 2023 (Impact factor : 3.2, Q1)
- SP Choi†, SW Park, **SJ Kang**, SK Lim, MS Kwon, HJ Choi, T Chun\*, Monitoring mRNA expression patterns in macrophages in response to two different strains of probiotics, *Food Science of Animal Resources* 43 (4), 703, 2023 (Impact factor : 3.7, Q2)
- CY Choi†, CH Lee, J Yang, **SJ Kang**, IB Park, SW Park, NY Lee, HB Hwang, HS Yun, T Chun\*, Efficacies of potential probiotic candidates isolated from traditional fermented Korean foods in stimulating immunoglobulin A secretion, *Food Science of Animal Resources* 43 (2), 346, 2023 (Impact factor : 3.7, Q2)
- SP Choi†, YC Choi, J Yang, CY Choi, CH Lee, **SJ Kang**, IB Park, T Chun\*, Monitoring mRNA transcription of genes involved in early pregnancy from endometrium and peripheral blood mononuclear cells of pregnant pigs with different parity, *Reproduction in Domestic Animals* 53 (6), 1594–1599, 2018 (Impact factor : 1.7, Q3)
- CY Choi†, YC Choi, IB Park, CH Lee, **SJ Kang**, T Chun\*, The ORF5 protein of porcine circovirus type 2 enhances viral replication by dampening type I interferon expression in porcine epithelial cells, *Veterinary Microbiology* 226, 50–58, 2018 (Impact factor : 2.7, Q1)

#### Patents

##### Registration

- T Chun, J Yang, CH Lee, **SJ Kang**, IB Park, Chimeric antigen receptor specifically binding to CD38 and use thereof, *Korean Intellectual Property Office*, Reg. No. 10-2231284, granted March 17, 2021
- T Chun, J Yang, CH Lee, **SJ Kang**, IB Park, Chimeric antigen receptor specifically binding to VLA-4 and use thereof, *Korean Intellectual Property Office*, Reg. No. 10-2231285, granted March 17, 2021

##### Pending

- **SJ Kang**, H Shin, Method and system for classifying intrinsically disordered proteins based on amino acid sequences using quantum neural networks, *Korean Intellectual Property Office*, 10-2025-0078871, June 2025

- T Chun, IB Park, **SJ Kang**, HJ Lee, Regulation of BCL11A expression by LDB1 transcriptional regulator and uses thereof, *Korean Intellectual Property Office*, 10-2025-0010845, January 2025
- T Chun, IB Park, **SJ Kang**, HJ Lee, Regulation of PLZF expression by LDB1 transcriptional regulator and uses thereof, *Korean Intellectual Property Office*, 10-2025-0010844, January 2025

## Certifications

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- **Big Data Analysis Engineer**, *Korea Data Agency*, 2025, BAE-010002862
- **ADsP (Advanced Data Analytics Semi-Professional)**, *Korea Data Agency*, 2024, ADsP-040004911

## Research Projects

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13. **Q2431261** 02/2025 – 12/2025  
**Title:** Safety evaluation of viral vector-based gene therapy with respect to non-specific genome insertion  
**Source:** Ministry of Food and Drug Safety  
**Role:** Research Assistant (*PI: Prof. Taehoon Chun*)
  - **Vector engineering** (Lentiviral/AAV) and **in vivo** administration
  - **WGS/NGS library construction** for genomic integration profiling
  - **Statistical analysis** of insertion site distribution and toxicity correlations
12. **Q2515581** 05/2025 – 12/2025  
**Title:** Optimization of HLA gene editing for the development of allogeneic stem cell therapy  
**Source:** Korea National Institute of Health  
**Role:** Research Assistant (*PI: Prof. Dongho Geum*)
  - **Hypoimmunogenic iPSC generation** via CRISPR-Cas9 HLA editing
  - **Directed differentiation** into immune-evasive cardiomyocytes and T cells
  - **In vivo engraftment validation** using humanized murine models
11. **R2222761** 09/2022 – 02/2025  
**Title:** Regulation of immune cell differentiation by chromatin 3D structural changes  
**Source:** National Research Foundation of Korea  
**Role:** Research Assistant (*PI: Prof. Taehoon Chun*)
  - **Grant Writing:** Multi-omics pipeline structuring and statistical planning
  - **Chromatin 3D structural analysis** mediated by LDB1 looper
  - **Multi-omics integration** (RNA/ATAC/ChIP-seq) for enhancer–promoter mapping
  - **In vivo validation** using LDB1-floxed mouse models
10. **R2212861** 04/2022 – 12/2023  
**Title:** Development of universal SARS-CoV-2 and sarbecovirus vaccine candidate using NDV vector platform  
**Source:** Korea Health Industry Development Institute  
**Role:** Research Assistant (*PI: Prof. Kisoan Kim*)
  - **Grant Writing:** Proposal submission and **AlphaFold application strategy**
  - **Structural epitope modeling** using AlphaFold for cross-protection
  - **Immunogenicity evaluation** in viral challenge animal models
  - **Immune profiling** via ELISA and flow cytometry
9. **R2131151** 02/2022 – 12/2025  
**Title:** Development of cellular immune evaluation technology for viral vector-based genetic vaccines  
**Source:** Ministry of Food and Drug Safety  
**Role:** Research Assistant (*PI: Prof. Taehoon Chun*)
  - **Candidate selection** for Adenovirus/Vaccinia-based platforms
  - **Assay development** for humoral and cellular immunity assessment
  - **Safety profiling** including cytotoxicity and organ-specific toxicity
8. **R1923081** 09/2019 – 02/2022  
**Title:** Mechanistic study of macrophage polarization regulated by chromatin remodeling factor Phc2

**Source:** National Research Foundation of Korea

**Role:** Research Assistant (*PI: Prof. Taehoon Chun*)

- **Transcriptomic profiling** of macrophage polarization states
- **Differential expression analysis** (DEG) and pathway enrichment
- **Regulatory network identification** of Phc2-mediated activation

7. R1727591 01/2018 – 12/2020  
**Title:** Development of DNA markers to enhance resistance against chronic wasting diseases  
**Source:** Rural Development Administration  
**Role:** Research Assistant (*PI: Prof. Taehoon Chun*)
  - **SNP genotyping** and sequence analysis for resistance markers
  - **Pipeline development** using Python/Biopython for variant detection
6. R1715822 03/2018 – 06/2022  
**Title:** Development of novel immune cell therapy targeting multiple myeloma  
**Source:** National Research Foundation of Korea  
**Role:** Research Assistant (*PI: Prof. Taehoon Chun*)
  - **CAR construct engineering** and optimization via molecular cloning
  - **Binder selection** for high affinity and specificity
  - **Kinetics analysis** ( $K_d$ ) using **surface plasmon resonance (SPR)**
5. Q2029461 01/2021 – 06/2023  
**Title:** Production and efficacy evaluation of novel recombinant immunotherapeutics  
**Source:** Immunological Designing Lab Co., Ltd.  
**Role:** Research Assistant (*PI: Prof. Taehoon Chun*)
  - **Recombinant protein production** and purification
  - **Affinity characterization** via SPR and ELISA
4. Q2015891 06/2020 – 02/2021  
**Title:** Screening of immune-enhancing probiotics using immunodeficient mouse models  
**Source:** CJ CheilJedang & BLOSSOM PARK  
**Role:** Research Assistant (*PI: Prof. Taehoon Chun*)
  - **In vivo screening** and cytokine profiling (Flow cytometry/qPCR)
  - **Statistical evaluation** of immune enhancement using R
3. Q1815701 06/2018 – 11/2018  
**Title:** Determination of dosing period and amount of probiotics for alleviating allergic rhinitis  
**Source:** CJ CheilJedang  
**Role:** Research Assistant (*PI: Prof. Taehoon Chun*)
  - **Th1/Th2 cytokine profiling** via ELISA and gene expression
  - **Dosing optimization modeling** for immune regulation
2. PJ013271012018 03/2018 – 12/2020  
**Title:** Development of genetic markers associated to the resistance to PMWS  
**Source:** Rural Development Administration  
**Role:** Research Assistant (*PI: Prof. Taehoon Chun*)
  - **Comparative genomics** for resistance marker identification
  - **Variant detection pipeline** using BLAST and Biopython
1. Q1613841 07/2016 – 04/2017  
**Title:** Selection and *in vivo* efficacy verification of probiotics with enhanced IgA secretion  
**Source:** CJ CheilJedang  
**Role:** Research Assistant (*PI: Prof. Taehoon Chun*)
  - **Mucosal IgA quantification** in murine models
  - **Immune gene validation** using qPCR

## Book chapters

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### International Publications

- *SJ Kang, 2026 Science Trends, Chapter: Multi-omics, Kindle, ISBN 9798266588004, September 2025*
- *SJ Kang, 2025 Science Trends, Chapter: Artificial Intelligence, Kindle, ISBN 979-8307079270, January 2025*

### Domestic Publications

- *SJ Kang, 2026 생명과학트렌드, 챕터: 멀티오믹스(2026 Life Science Trends, Chapter: Multi-omics), Pubple, ISBN 9788924169034, August 2025*
- *SJ Kang, 2026 과학트렌드, 챕터: 개인 맞춤형 치료(2026 Science Trends: Personalized Medicine), Pubple, ISBN 9788924169041, August 2025*
- *SJ Kang, 2025 과학트렌드, 챕터: 인공지능(2025 Science Trends, Chapter: Artificial Intelligence), Pubple, ISBN 9788924143621, December 2024*

## Technical Skills

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**Wet Lab:** Flow cytometry (multicolor panel design & sorting), MACS, ELISA, qPCR, SPR (Biacore), Western blot, Molecular cloning, CRISPR-Cas9 editing, Combinatorial library construction, Yeast surface display, Recombinant antibody/nanobody expression & purification (affinity, SEC), NGS library preparation, Lentiviral/AAV vector production, Cell culture, Mouse models (tumor, immunization)

**Computational:** scRNA-seq analysis, scATAC-seq analysis, ChIP-seq & Hi-C (HOMER, MACS2, deepTools, cooler), QIIME2, MD simulation (GROMACS), AlphaFold2/3, ProteinMPNN, Rosetta, ESM-2, Deep learning (PyTorch), Machine learning, Python, R

## References

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### Prof. Taehoon Chun

Department of Biotechnology, Korea University  
*Relationship: Ph.D. Advisor*  
tchun@korea.ac.kr

### Prof. Jong-Hoon Kim

Department of Biotechnology, Korea University  
*Relationship: Collaborator*  
jhkim@korea.ac.kr

### Prof. Joonbeom Bae

Department of Biotechnology, Korea University  
*Relationship: Former Labmate & Collaborator*  
joonbeombae@korea.ac.kr